Shape and Space 2 — Pythagoras' theorem · Circle theorems

Calculator allowed. Give lengths to 3 s.f. unless exact. For every angle, give the circle theorem or angle fact you used. Marks in brackets. ★ = harder question.

A · PYTHAGORAS' THEOREM

1 In each right-angled triangle, find the missing side. [4]

a The two shorter sides are 9 cm and 12 cm. Find the hypotenuse.

b The hypotenuse is 13 cm and one shorter side is 5 cm. Find the other side.

2 A ramp 3 m long leads up to a step 30 cm high. How far from the step does the ramp start? Give your answer in metres to the nearest centimetre. [3]

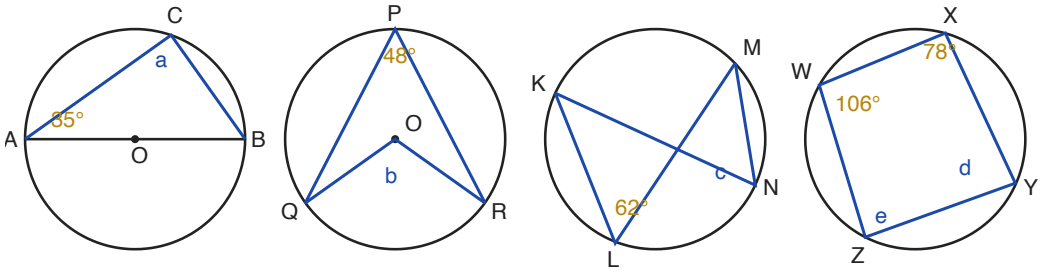
3 Is a triangle with sides 7 cm, 24 cm and 25 cm right-angled? Show how you decide. [2]

4 Find the distance between the points $A(1,\ 2)$ and $B(7,\ 10)$. [3]

5 A 6.5 m ladder leans against a wall and reaches 6 m up it. How far is the foot of the ladder from the wall? [2]

B · CIRCLE THEOREMS

6 In each diagram O is the centre of the circle. Work out the angles marked with letters, giving a reason for each. [9]



Left: AB is a diameter, angle $CAB = 35°$; find a (angle ABC). Second: angle $QPR = 48°$; find b (angle QOR). Third: chords KN and LM; angle $KLM = 62°$; find c (angle KNM). Right: cyclic quadrilateral $WXYZ$ with $W = 106°$, $X = 78°$; find d (angle Y) and e (angle Z).

7 A and B lie on a circle with centre O, and angle $AOB = 130°$. C is a point on the major arc. [3]

a Work out angle ACB.

b Work out angle OAB.

8 Find the area of an equilateral triangle with perimeter 30 cm. [3]

Total: 29 marks

Check: hypotenuse is opposite the right angle — add for it, subtract for a shorter side · same units before squaring · quote the theorem by name: semicircle, centre = 2 × circumference, same segment, cyclic quadrilateral.